CHEMICAL REACTIONS AND EQUATIONS

📖 Definitions 📝

- **Chemical Change:** A permanent change in which the chemical composition of a substance is changed, and one or more new substances with different chemical compositions and different properties are formed. The change is not easily reversible. 🔄
- **Chemical Reaction:** The process of breaking the chemical bonds of the reacting substances (reactants) and making new bonds to form new substances (products). ⚗️
- **Chemical Bond:** The force that holds the atoms of a molecule together, as in a compound. 🤝
- **Effervescence:** The formation of gas bubbles in a liquid during a reaction. 🫧

💡 The "Why" & Extra Key Facts 🧠

- **Why brown bottles?** Solutions of silver nitrate and hydrogen peroxide are kept in brown bottles in the laboratory because they decompose in the presence of light. Silver nitrate becomes black, while hydrogen peroxide changes to water and oxygen. 🟤
- **Enzymes (Positive Catalysts):** Enzymes are proteins that act as positive catalysts in biochemical reactions. During digestion, complex food material gets broken down into simpler substances in two to three hours in the presence of enzymes; otherwise, it would take months! 🧬
- **Promoters:** Substances that influence the rate of a chemical reaction by improving the efficiency of the catalyst. (Note: A promoter alone cannot increase the rate of reaction). 🚀
- **Negative Catalysts (Inhibitors):** A catalyst employed to retard (slow down) the rate of a reaction. 🛑

🔄 Characteristics of a Chemical Change 📌

1. A chemical change is a permanent and irreversible change. 🔒

2. The composition of the substance(s) undergoing a chemical change changes. 🧪
3. A chemical change is either exothermic (where heat is evolved) or endothermic (where heat is absorbed). 🔥 ❄️
4. The total mass of the reactants and products in a chemical change remains the same (Law of conservation of mass). ⚖️

⚡ Conditions Necessary for a Chemical Reaction ⚙️

(For particles to collide and react, one or more of these conditions is necessary)

- Mixing (close contact) 🤝
- Solution (molten or aqueous state) 💧
- Heat ♨️
- Light 🌞
- Electricity 🔌
- Pressure 🧱
- Catalyst ⏩
- Sound Energy 🔊

📋 Characteristics of Chemical Reactions 🔍

- Formation of a new substance ✨
- Change of colour 🎨
- Evolution of gas 💨
- Formation of precipitates (insoluble solid substances) ⬇️
- Change in temperature (exothermic or endothermic) 🌡️
- Change of state 🧊 💧
- Change in odour 👃

🧪 Reactions Master List (Categorized by Condition) 🚀

1. Conditions: Mixing (Close Contact) 🤝

- *Word Reaction:* Iodine and phosphorus react explosively when brought into close contact. 💥
- *Reaction:* $Pb(NO_3)_2(s) + 2KI(s) \rightarrow 2KNO_3(s) + PbI_2(s)$
    - Lead nitrate (white) + potassium iodide (white) → lead iodide (yellow).

2. Conditions: Solution 💧

- *Word Reaction:* Oxalic acid crystals and sodium carbonate react in water solution only. 🥣
- *Reaction:* $NaCl(aq) + AgNO_3(aq) \rightarrow AgCl\downarrow + NaNO_3(aq)$
    - Sodium chloride (solution state) + silver nitrate (solution state) → silver chloride (white ppt.) + sodium nitrate.

3. Conditions: Heat (Δ) ♨️

- *Reaction:* $CuCO_3(s) \xrightarrow{\Delta} CuO(s) + CO_2(g)$
    - Copper carbonate (green powder) → copper oxide (black) + carbon dioxide.
- *Reaction:* $2Pb(NO_3)_2 \xrightarrow{\Delta} 2PbO + 4NO_2 + O_2$
    - Lead nitrate (white) → lead monoxide (yellow residue) + nitrogen dioxide (brown gas) + oxygen (colourless gas).

4. Conditions: Light (Photochemical Reactions) 🌞

- *Reaction:* $6CO_2 + 12H_2O \xrightarrow{\text{Light}} C_6H_{12}O_6 + 6O_2 + 6H_2O$
    - (Plants form glucose). 🌱
- *Reaction:* $H_2 + Cl_2 \xrightarrow{\text{Sunlight}} 2HCl$
    - (Hydrogen and chlorine react).
- *Reaction:* $Cl_2 + H_2O \rightarrow HCl + HClO$ (then) $2HClO \xrightarrow{\text{Sunlight/UV radiation}} 2HCl + O_2$
    - (Chlorine water exposed to sunlight).
- *Reaction:* $2AgNO_3 \xrightarrow{\text{sunlight}} 2Ag + 2NO_2 + O_2$

- Silver nitrate becomes black.

- *Reaction:* $2H_2O_2 \xrightarrow{\text{sunlight}} 2H_2O + O_2$
  - Hydrogen peroxide decomposes.

5. **Conditions: Electricity (Electrochemical Reactions)** 🔌
   - *Reaction:* $2H_2O \xrightarrow{\text{electricity}} 2H_2 + O_2$
     - (Electrolysis of acidulated water).
   - *Reaction:* $2NaCl \xrightarrow{\text{electricity}} 2Na + Cl_2$
     - (Molten sodium chloride).
6. **Conditions: Pressure** 🧱
   - *Reaction:* $HgCl_2 + 2KI \rightarrow HgI_2 + 2KCl$
     - Mercuric chloride + potassium iodide (crushed together at high pressure) → mercuric iodide (scarlet-coloured).
7. **Conditions: Catalyst** ⏩ *(All requested reactions included here!)*
   - **General Catalyst Reactions:**
     - *Reaction 1:* $2KClO_3 \xrightarrow{MnO_2} 2KCl + 3O_2$
       - Potassium chlorate decomposes. Manganese dioxide acts as a catalyst so decomposition begins at a much lower temperature ($300°C$), and manganese dioxide remains unaffected.
     - *Reaction 2:* $4NH_3 + 5O_2 \xrightarrow[800°C]{Pt} 4NO + 6H_2O$
       - Ammonia reacts with oxygen to produce nitric oxide and water vapour in the presence of platinum, a catalyst.
   - **Positive Catalyst:**
     - *Reaction:* $2H_2O_2 \xrightarrow{MnO_2} 2H_2O + O_2$
       - The rate of decomposition of hydrogen peroxide increases in the presence of manganese dioxide (positive catalyst).

- **Promoters:**
    - *Reaction:* $N_2 + 3H_2 \rightleftharpoons 2NH_3$ *(with Iron + Mo written over the arrow)*
        - In the manufacture of ammonia, iron acts as a catalyst and molybdenum (Mo) as a promoter to increase efficiency.
- **Negative Catalyst:**
    - *Word Reaction 1:* Phosphoric acid retards the rate of decomposition of hydrogen peroxide. 📉
    - *Word Reaction 2:* The rate of oxidation of chloroform decreases in the presence of alcohol. 📉

8. Conditions: Sound Energy 🔊

- *Reaction:* $C_2H_2 \xrightarrow{\text{Sound energy}} 2C + H_2$
    - Acetylene breaks up into carbon and hydrogen by absorbing sound energy.

🎨 **Reactions Showing Characteristic Changes** 🔍

- **Formation of a New Substance:**
    - $2Na + 2H_2O \rightarrow 2NaOH + H_2$
        - [sodium] + [water] → [sodium hydroxide] + [hydrogen].
- **Change of Colour:**
    - $Fe + CuSO_4(aq) \rightarrow FeSO_4 + Cu$
        - [iron] + [blue solution] → [light green solution] + [brown red deposit].
    - $Cu(s) + 2H_2SO_4(aq) \rightarrow CuSO_4(aq) + 2H_2O(l) + SO_2(g)$
        - [brown/pink] + [conc.] → [blue].
- **Evolution of Gas (Effervescence):**
    - $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2 \uparrow$
        - [zinc] + [dil. sulphuric acid] → [zinc sulphate] + [hydrogen].
    - $CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g) \uparrow$

        - Calcium carbonate + dilute hydrochloric acid evolves carbon dioxide.

- **Formation of Precipitates:**
    - $AgNO_3(aq) + NaCl(aq) \rightarrow AgCl \downarrow + NaNO_3(aq)$
        - [silver nitrate solution] + [sodium chloride solution] → [silver chloride] (curdy white ppt) + [sodium nitrate solution].
    - $FeSO_4(aq) + 2NaOH(aq) \rightarrow Fe(OH)_2 \downarrow + Na_2SO_4(aq)$
        - [ferrous sulphate] + [sodium hydroxide] → [ferrous hydroxide] (dirty green ppt) + [sodium sulphate].
- **Change in Temperature:**
    - $CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq) + Heat$
        - [quick lime] → [slaked lime] (Exothermic).
    - $C(s) + 2S(s) \rightarrow CS_2(l)$
        - Carbon + sulphur (on heating together) (Endothermic).
- **Change of State:**
    - $NH_3(g) + HCl(g) \rightleftharpoons NH_4Cl(s)$
        - Ammonia gas + hydrogen chloride gas → solid ammonium chloride.
- **Change in Odour:**
    - $FeS + 2HCl \rightarrow FeCl_2 + H_2S$
        - [iron sulphide] + [hydrochloric acid] → [hydrogen sulphide gas] (rotten egg smell). 🥚🍽️

📖 Definitions & Classifications 📝

- Chemical reactions have been classified into four different types: 1. Direct combination (or synthesis), 2. Decomposition, 3. Displacement, 4. Double decomposition.
- **Direct Combination (or Synthesis):** A reaction in which two or more substances combine together to form a single substance is called a

combination reaction or synthesis ($A + B \rightarrow AB$). The reactants may be elements or compounds but the product is always a compound.

- **Decomposition:** The chemical reaction in which a compound splits into two or more simpler substances (elements or compounds) is called decomposition reaction ($AB \rightarrow C + D + \cdots$). Decomposition is the breaking up of a compound either into elements or simpler compounds, such that these products do not recombine to form the original compound.
- **Thermal Decomposition:** A decomposition reaction that is brought about by heat without any recombination on cooling is known as thermal decomposition. The symbol Δ (delta) is used to signify that heat has caused the reaction.

💡 The "Why" & Extra Key Facts 🧠

- **Whitewashing:** Slaked lime is used for white washing. When a solution of slaked lime is applied to the walls, calcium hydroxide reacts slowly with carbon dioxide present in the atmosphere to form a thin layer of calcium carbonate on the walls of the building. The layer of calcium carbonate formed gives a shiny finish. Reaction: $Ca(OH)_2(aq) + CO_2(g) \rightarrow CaCO_3(s) + H_2O(l)$.
- **Bases vs. Alkalis:** Metallic oxides (Na_2O, ZnO, CaO) are called bases. Bases that dissolve in water are called alkalis (e.g., $NaOH, KOH, NH_4OH$, and $Ca(OH)_2$).
- **Acidic Anhydrides:** Non-metallic oxides which dissolve in water form acids. Non-metallic oxides which react with water to form acids are known as acidic anhydrides (e.g., SO_2 and P_2O_5 are acidic anhydrides of sulphurous acid and phosphoric acid respectively).
- **Ferrous Sulphate Crystals:** Ferrous sulphate crystals are actually ferrous sulphate heptahydrate ($FeSO_4 \cdot 7H_2O$). These crystals are green in colour, on heating in a test tube they change to white and then a brown solid (Fe_2O_3) is formed. The gas coming out is a mixture of sulphur dioxide (SO_2) and sulphur trioxide (SO_3) having a suffocating and pungent smell of burning sulphur. Sulphur dioxide gas turns acidified potassium dichromate paper green.

🧪 Direct Combination (Synthesis) Reactions ➕

1. Two Elements Combine to Form a Compound: 🤝

- *Iron & Sulphur:* The reaction between iron and sulphur to form iron (II) sulphide requires heat to start, but continues exothermally.
    - $Fe(s) + S(s) \xrightarrow{heat} FeS(s)$.
- *Iron & Chlorine:* Iron (III) chloride, a brown compound, is prepared when dry chlorine is passed over heated iron.
    - $2Fe(s) + 3Cl_2(g) \rightarrow 2FeCl_3(s)$.
- *Nitrides Formation:* Burning magnesium reacts with nitrogen to form magnesium nitride.
    - $3Mg + N_2 \rightarrow Mg_3N_2$.
- *Ammonia Formation:* Hydrogen reacts with nitrogen under pressure above 200 atmospheres and temperature $450°C$ in presence of iron as a catalyst to form ammonia.
    - $N_2 + 3H_2 \rightleftharpoons 2NH_3$.
- *Metals & Non-metals with Oxygen:*
    - $4Na + O_2(g) \rightarrow 2Na_2O(s)$
    - $4Fe + 3O_2(g) \xrightarrow{\Delta} 2Fe_2O_3(s)$
    - $3Fe + 2O_2(g) \xrightarrow{\Delta} Fe_3O_4(s)$
    - $2Hg + O_2(g) \xrightarrow{\Delta} 2HgO(s)$
    - $N_2 + O_2(g) \xrightarrow{3000°C} 2NO(g)$
    - $4P + 5O_2(g) \xrightarrow{\Delta} 2P_2O_5(s)$
    - $C(s) + O_2(g) \xrightarrow{\Delta} CO_2(g)$
- *Experiments:*
    - Black lead sulphide reacts with oxygen on heating to form white lead sulphate. $PbS + 2O_2 \rightarrow PbSO_4$.

- Magnesium ribbon burns over a flame with a dazzling white light, forming magnesium oxide. $2Mg + O_2 \rightarrow 2MgO$.

2. An Element and a Compound Combine: 🔗

- Carbon monoxide (compound) burns in the presence of oxygen (element) to form carbon dioxide.
  - $2CO(g) + O_2(g) \xrightarrow{heat} 2CO_2(g)$.
- Sulphur dioxide and oxygen combine under certain conditions to form sulphur trioxide.
  - $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$.

3. Two or More Compounds Combine: 🤝🔗

- Ammonia and hydrogen chloride combine to form ammonium chloride.
  - $NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$.
- Soluble metal oxides react with water to form their hydroxides.
  - $K_2O + H_2O \rightarrow 2KOH$ (Caustic potash).
  - $Na_2O + H_2O \rightarrow 2NaOH$ (Caustic soda).
- When calcium oxide (quick lime) combines with water, a vigorous reaction takes place with the liberation of a large amount of heat (exothermic reaction) and calcium hydroxide (or slaked lime) is formed.
  - $CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq)$.
- Non-metallic oxides dissolve in water to form acids.
  - $SO_2 + H_2O \rightarrow H_2SO_3$ (sulphurous acid).
  - $P_2O_5 + 3H_2O \rightarrow 2H_3PO_4$ (phosphoric acid).
- Photosynthesis.
  - $6CO_2 + 6H_2O \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} C_6H_{12}O_6 + 6O_2$.

💥 **Decomposition Reactions** 🪓

(Decomposition may occur in the presence of heat, light, or by the passage of an electric current.)

1. **A Compound Breaks Up Into Two or More Elements:**

- *Heat:* Mercuric oxide, when heated, decomposes to form mercury and oxygen.
    - $2HgO(s) \xrightarrow{\Delta} 2Hg(l) + O_2(g)$.
- *Light:* When silver chloride (white) is exposed to sunlight, it undergoes photochemical decomposition forming a grey metal, silver and a greenish yellow gas, chlorine.
    - $2AgCl(s) \xrightarrow{\text{sunlight}} 2Ag(s) + Cl_2(l)$.
- *Light:* Silver bromide (light yellow) decomposes to silver (grey) and bromine (brown).
    - $2AgBr(s) \rightarrow 2Ag(s) + Br_2(g)$.
- *Electricity:* When electric current is passed through acidulated water, it decomposes into hydrogen and oxygen.
    - $2H_2O(l) \xrightarrow{\text{electric current}} 2H_2(g) + O_2(g)$.

2. **A Compound Breaks Up to Form Elements and Compounds:**

- Potassium nitrate, on heating, decomposes to produce a compound, potassium nitrite, and an element, oxygen.
    - $2KNO_3(s) \xrightarrow{\text{heat}} 2KNO_2(s) + O_2(g)$.

3. **A Compound Breaks Up to Form Two or More New Compounds:**

- Calcium carbonate on strong heating decomposes to form calcium oxide and carbon dioxide.
    - $CaCO_3(s) \xrightarrow[1000^\circ C]{\text{heat}} CaO(s) + CO_2(g)$.
- Ferrous sulphate undergoes thermal decomposition to give ferric oxide, sulphur dioxide and sulphur trioxide.
    - $2FeSO_4(s) \xrightarrow{\text{heat}} Fe_2O_3(s) + SO_2(g) + SO_3(g)$ ([white] -> [brown]).

🔥 **Thermal Decomposition of Metal Compounds** 🧱

(Decomposition by heat is based on reactivity; compounds of reactive metals are more stable and do not decompose easily, while less reactive metal compounds decompose readily.)

1. **Metal Hydroxides:**
   - *K, Na:* Stable to heat.
   - *Ca, Mg, Al, Zn, Fe, Pb, Cu:* Decompose on heating to form metal oxide and water vapour.
     - $Ca(OH)_2 \rightarrow CaO + H_2O$
     - $Zn(OH)_2 \rightarrow ZnO + H_2O$
     - $Pb(OH)_2 \rightarrow PbO + H_2O$
     - $Cu(OH)_2 \rightarrow CuO + H_2O$
   - *Hg, Ag:* Yield metal, oxygen and water vapour.
     - $4AgOH \rightarrow 4Ag + O_2 + 2H_2O$
2. **Metal Carbonates:**
   - *K, Na:* Stable to heat and soluble in water.
   - *Ca, Mg, Al, Zn, Fe, Pb, Cu:* Decompose on heating with decreasing vigour to form metal oxide and carbon dioxide.
     - $MgCO_3 \rightarrow MgO + CO_2$
     - $ZnCO_3 \rightarrow ZnO + CO_2$
     - $CuCO_3 \rightarrow CuO + CO_2$
   - *Hg, Ag:* Decompose on heating to form metal, oxygen and carbon dioxide.
     - $2Ag_2CO_3 \rightarrow 4Ag + O_2 + 2CO_2$
3. **Metal Bicarbonates (Metal Hydrogen Carbonate):**
   - Decompose to give metal carbonates, water vapour and carbon dioxide.
     - $2NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + H_2O + CO_2$
     - $Ca(HCO_3)_2 \xrightarrow{\Delta} CaCO_3 + H_2O + CO_2$
     - $Mg(HCO_3)_2 \xrightarrow{\Delta} MgCO_3 + H_2O + CO_2$

📖 Definitions 📝

- **Dehydration Reaction:** A reaction in which a hydrated salt loses molecules of water of crystallisation. 💧
- **Reversible Reactions:** A group of reactions in which the direction of chemical change can be reversed by changing the conditions under which the reaction is taking place (indicated by the sign "⇌"). 🔁
- **Thermal Dissociation:** A reversible decomposition reaction brought about only by heat. 🔥
- **Displacement Reaction:** A chemical change in which a more active element displaces a less active element from its salt solution ($AB + C \rightarrow AC + B$). 🔀
- **Double Decomposition:** A type of chemical change in which two compounds in a solution react to form two new compounds by mutual exchange of radicals ($AB + CD \rightarrow AD + CB$). It has two types: precipitation and neutralization. 🔀🔀
- **Precipitation Reaction:** A chemical reaction in which two compounds in their aqueous state react to form an insoluble salt (a precipitate) as one of the products. ⬇️
- **Neutralization:** The reaction between an acid and a base that forms salt and water only. It happens because the hydrogen ion (H^+) from the acid combines with the hydroxyl ion (OH^-) from the base to form water ($H^+ + OH^- \rightarrow H_2O$). **Note: Double decomposition reactions in aqueous acid and base solutions result in the formation of a neutral salt and are therefore considered neutralization reactions.** ⚖️

💡 The "Why" & Extra Key Facts 🧠

- **Decomposition in our body:** Digestion of food is a decomposition reaction. Starch decomposes into glucose and sugar; proteins undergo decomposition to form amino acids; fats and oils are decomposed to fatty acids and finally oxidized by respiration into carbon dioxide and water. Reaction: Starch $\xrightarrow{\text{Enzymes}}$ Glucose $\xrightarrow{[O]} CO_2 + H_2O +$ Energy. 🍎🧬

- **Reactivity Series Rules:** Less reactive elements cannot displace more active elements. For example, copper cannot displace zinc from its salt solution ($ZnSO_4 + Cu \rightarrow$ No product). 🚫
- **Testing for SO_4^{2-} (Sulphate) ions:** Any soluble sulphate reacting with Barium Chloride ($BaCl_2$) will form a white precipitate of Barium Sulphate ($BaSO_4$). 🧪
- **Testing for Cl^- (Chloride) ions:** Any chloride soluble in water reacting with Silver Nitrate ($AgNO_3$) forms a white curdy precipitate of Silver Chloride ($AgCl$). 🥛
- **Testing for CO_2 gas:** Passing carbon dioxide through lime water turns it milky due to the formation of insoluble calcium carbonate. If passed in excess, the milkiness disappears due to the formation of soluble calcium hydrogen carbonate. 🫧
- **Everyday Neutralization:**
    - **Bee stings:** Formic acid enters the skin causing pain; relieved by rubbing slaked lime or baking soda (both basic) to neutralize the acid. 🐝
    - **Stomach pain:** Caused by excess HCl secretion; relieved by milk of magnesia or sodium hydrogen carbonate (both basic). 💊
    - **Acid spills:** Neutralized on clothes with ammonia solution, or on skin with sodium hydrogen carbonate paste. 🥼
    - **Acidic soil:** Slaked lime is added to neutralize excess acid for better crops. 🌱

🧪 Reactions Master List 🚀

1. Decomposition of Metal Nitrates (By Heat) 🔥

- *K, Na (Melt and give nitrite + oxygen):*
    - $2KNO_3 \rightarrow 2KNO_2 + O_2$
- *Ca, Mg, Al, Zn, Fe, Pb, Cu (Form metal oxide + nitrogen dioxide + oxygen):*
    - $2Ca(NO_3)_2 \rightarrow 2CaO + 4NO_2 + O_2$
    - $2Zn(NO_3)_2 \rightarrow 2ZnO + 4NO_2 + O_2$

- $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 \uparrow + O_2 \uparrow$ (Lead nitrate crystals melt, give reddish brown gas NO_2, and leave a yellow solid residue of lead monoxide)
- $2Cu(NO_3)_2 \rightarrow 2CuO + 4NO_2 + O_2$

- *Hg, Ag (Form metal + nitrogen dioxide + oxygen):*
  - $2AgNO_3 \xrightarrow{\Delta} 2Ag + 2NO_2 + O_2$
- *Other Decomposition Experiments:*
  - $ZnCO_3 \rightarrow ZnO + CO_2 \uparrow$ (Zinc carbonate heated; residue is yellow when hot, white on cooling).
  - $(NH_4)_2Cr_2O_7 \rightarrow Cr_2O_3 + 4H_2O \uparrow + N_2 \uparrow$ (Orange ammonium dichromate decomposes with flashes of light into a green solid chromium oxide).

2. Dehydration & Reversible Reactions / Thermal Dissociation 🔁

- *Dehydration:* $CuSO_4 \cdot 5H_2O(s) \rightarrow CuSO_4(s) + 5H_2O(g)$ (Blue hydrated salt turns into white anhydrous salt).
- *Reversible (Adding water back):* $CuSO_4(s) + 5H_2O(l) \rightarrow CuSO_4 \cdot 5H_2O(s)$ (White anhydrous turns back to blue).
- *Reversible (Iron and steam):* $3Fe + 4H_2O \rightleftharpoons Fe_3O_4 + 4H_2$.
- *Thermal Dissociation (Ammonium chloride):* $NH_4Cl \rightleftharpoons NH_3 \uparrow + HCl \uparrow$ (Solid decomposes to two colourless gases, then recombines as a white sublimate on cooling).
- *Thermal Dissociation (Nitrogen tetraoxide):* $N_2O_4 \backslash xrightleftharpoons[cool]{heat} 2NO_2$ (Original compound changes to a reddish brown gas on heating).

3. Displacement Reactions 🔀

- *Metals from Water:*
  - $2K + 2H_2O \rightarrow 2KOH + H_2$ (cold water)
  - $2Na + 2H_2O \rightarrow 2NaOH + H_2$ (cold water)
  - $Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$ (cold water)

- $Mg + H_2O \rightarrow Mg(OH)_2 + H_2$ (requires heat)

- *Metals from Acids:*
  - $2K + 2HCl \rightarrow 2KCl + H_2$
  - $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
  - $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$
  - $Mg + H_2SO_4 \rightarrow MgSO_4 + H_2$ (Brisk effervescence, hydrogen burns with a pop sound)
- *Metals from Salt Solutions:*
  - $CuSO_4(aq) + Zn \rightarrow ZnSO_4 + Cu \downarrow$ (Blue solution fades to colourless; reddish brown copper particles settle).
- *Non-metals:*
  - $2KI + Cl_2 \rightarrow 2KCl + I_2$ (Colourless solution turns yellow brown as iodine is formed).

4. Double Decomposition (Precipitation & Neutralization) ⬇️ 💧

- *Precipitation (Forming an insoluble solid):*
  - $BaCl_2(aq) + Na_2SO_4(aq) \rightarrow BaSO_4(s) \downarrow + 2NaCl(aq)$ (white ppt)
  - $CuSO_4(aq) + H_2S(g) \rightarrow CuS(s) \downarrow + H_2SO_4(aq)$ (black ppt)
  - $FeCl_2(aq) + 2NaOH(aq) \rightarrow Fe(OH)_2 \downarrow + 2NaCl(aq)$ (green ppt)
  - $H_2SO_4(aq) + BaCl_2(aq) \rightarrow BaSO_4 \downarrow + 2HCl(aq)$ (white ppt, test for sulphate)
  - $NaCl(aq) + AgNO_3(aq) \rightarrow AgCl \downarrow + NaNO_3(aq)$ (white curdy ppt, soluble in ammonium hydroxide)
  - $AgNO_3(aq) + HCl(aq) \rightarrow AgCl \downarrow + HNO_3(aq)$ (white ppt, test for chloride)
  - $Pb(NO_3)_2(aq) + 2KI(aq) \rightarrow PbI_2 \downarrow + 2KNO_3(aq)$ (Thick bright yellow ppt, dissolves on heating and reappears on cooling)
- *With Evolution of a Gas:*

- $FeS(s) + H_2SO_4(aq) \rightarrow FeSO_4(aq) + H_2S \uparrow$ (offensive rotten egg smelling gas)
- $ZnS(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2S \uparrow$
- $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2 \uparrow$ (Brisk effervescence)
- $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 \downarrow + H_2O$ (Lime water turns milky)
- $CaCO_3(s) + H_2O + CO_2 \rightarrow Ca(HCO_3)_2(aq)$ (Milkiness disappears due to soluble product)

- *Neutralization (Acid + Base → Salt + Water):*
  - **Soluble Base (Alkali) + Acid:**
    - $NaOH + HCl \rightarrow NaCl + H_2O$
    - $KOH + HNO_3 \rightarrow KNO_3 + H_2O$
    - $2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$
    - $NH_4OH + HCl \rightarrow NH_4Cl + H_2O$
  - **Insoluble Base + Acid:**
    - $PbO + 2HNO_3 \rightarrow Pb(NO_3)_2 + H_2O$
    - $CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$
    - $Pb(OH)_2 + 2HNO_3 \rightarrow Pb(NO_3)_2 + 2H_2O$
    - $Cu(OH)_2 + H_2SO_4 \rightarrow CuSO_4 + 2H_2O$

📖 Definitions & Concepts 📝

- **Chemical Energy:** Every substance has a fixed amount of stored energy, which is in the form of potential energy, and is referred to as its chemical energy.
- **Energy Change in a Reaction:** A chemical reaction involves the breaking up of chemical bonds (absorbing energy) and the simultaneous formation of bonds (releasing energy). The total energy change equals the total energy released on the product side minus the total energy released on the reactant side. This energy can be in the form of heat, light, sound, or electricity.

- **Exothermic Reaction:** A chemical reaction in which heat (a form of energy) is given out. It causes a rise in temperature. In these reactions, the total energy content of the reactants is more than that of the products.
- **Endothermic Reaction:** A chemical reaction in which heat is absorbed. It causes a fall in temperature. These reactions cannot be sustained without a supply of energy from an external source. The total energy content of the products is more than that of the reactants.
- **Photochemical Reactions:** Certain chemical changes or reactions that occur with the absorption of light energy.
- **Electrochemical Reactions:** Reactions that occur with the absorption of electrical energy. They can be either exothermic (like a battery powering a device) or endothermic (like charging a battery or electrolysis).

💡 The "Why" & Extra Key Facts 🧠

- **Fuels:** Substances such as methane, which undergo combustion readily and give out a large amount of energy, are known as fuels.
- **Slaking of Lime:** When water is added to quicklime, a lot of heat energy is produced, boiling the water and making a hissing sound as alkaline calcium hydroxide (slaked lime) forms.
- **Respiration:** Our bodies need energy to stay alive, which we get from food. Carbohydrates are broken down into glucose. The combustion of this glucose with oxygen in our cells is an exothermic reaction known as respiration.
- **Other Exothermic Examples:** Rusting, disintegration of radioactive material, dynamite explosions, and the burning of coal, petrol, kerosene, or cooking gas are all exothermic.
- **Photosynthesis:** The green pigment chlorophyll is essential for photosynthesis because it traps energy from the sun, making it a photochemical reaction.
- **Other Endothermic Examples:** Electrolysis of water and extraction of iron ore are endothermic reactions.

🧪 Reactions Master List 🚀

1. Exothermic Reactions (Heat is Evolved) 🔥

- *Burning Carbon:* When carbon burns in oxygen to form carbon dioxide, a lot of heat is produced.
  - $C + O_2 \rightarrow CO_2 +$ Heat
- *Combustion of Natural Gas:* Methane's complete combustion produces carbon dioxide and water vapour.
  - $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$
- *Slaking of Lime:* Water added to quicklime forms slaked lime and heat.
  - $CaO + H_2O \rightarrow Ca(OH)_2 +$ Heat
- *Formation of Water:* Hydrogen burns in oxygen to form water and heat.
  - $2H_2 + O_2 \xrightarrow{\Delta} 2H_2O +$ Heat
- *Formation of Ammonia:* Nitrogen reacts with hydrogen (with iron catalyst at $450°C$ to $500°C$, >200 atm) to form ammonia.
  - $N_2 + 3H_2 \xrightarrow{\Delta} 2NH_3 +$ Heat
- *Neutralisation:* Acid and base react to give salt, water, and heat.
  - $NaOH + HCl \rightarrow NaCl + H_2O +$ Heat
- *Respiration:* Combustion of glucose.
  - $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O +$ Energy

2. Endothermic Reactions (Heat is Absorbed) ❄️

- *Formation of Carbon Disulphide:* Carbon heated with sulphur at high temperature forms liquid carbon disulphide.
  - $C + 2S \xrightarrow{\Delta} CS_2$
- *Formation of Nitric Oxide:* Nitrogen and oxygen heated to $3000°C$ form nitric oxide gas.
  - $N_2 + O_2 \underset{3000°C}{\overset{\text{Heat}}{\rightarrow}} 2NO(g)$

- *Decomposition of Calcium Carbonate:* Calcium carbonate heated to $1000°C$ decomposes to calcium oxide and carbon dioxide.
    - $CaCO_3(s) \xrightarrow[1000°C]{\text{heat}} CaO(s) + CO_2(g)$

3. Photochemical Reactions (Involving Light) 🌞

- *Photosynthesis:* Takes place in green leaves containing chlorophyll.
    - $6CO_2 + 12H_2O \xrightarrow[\text{sunlight}]{\text{chlorophyll}} C_6H_{12}O_6 + 6H_2O + 6O_2$
- *Decomposition of Silver Nitrate:* Breaks down in sunlight.
    - $2AgNO_3 \xrightarrow{\text{sunlight}} 2Ag + 2NO_2 + O_2$
- *(Note: Conversely, Light is Evolved during combustion)*: Magnesium ribbon produces light and heat.
    - $2Mg + O_2 \rightarrow 2MgO + \text{Light}$

4. Electrochemical Reactions (Involving Electricity) ⚡

- *Fused Potassium Chloride:* Passing current breaks it into charged particles.
    - $KCl \xrightarrow[\text{current}]{\text{electric}} K^+ + Cl^-$
- *Electrolysis of Acidulated Water:* Breaks into hydrogen and oxygen.
    - $2H_2O \xrightarrow[\text{current}]{\text{electric}} 2H_2 + O_2$

*GO*1. **(a) What is a chemical reaction?**

A chemical reaction is the process of breaking the chemical bonds of the reacting substances (reactants) and making new bonds to form new substances (products). 🧪

(b) State the conditions necessary for a chemical change or reaction to take place.

The conditions necessary are: mixing (close contact), solution, heat, light, electricity, pressure, presence of a catalyst, or sound energy. ⚙️

2. Define the following terms

(a) Chemical change

A chemical change is a permanent and irreversible change in which the chemical composition of a substance is changed, and one or more new substances with different chemical compositions and properties are formed. 🔄

(b) Chemical bond

A chemical bond is the force that holds the atoms of a molecule together, as in a compound. 🤝

(c) Effervescence

The formation of gas bubbles in a liquid during a reaction is called effervescence. 🫧

(d) Precipitate

A precipitate is an insoluble solid substance formed during certain chemical reactions, typically when two solutions are mixed. ⬇️

3. Give an example of a reaction where the following are involved

(a) Heat

$CuCO_3(s) \xrightarrow{\Delta} CuO(s) + CO_2(g)$ 🔥

(b) Light

$H_2 + Cl_2 \xrightarrow{\text{Sunlight}} 2HCl$ 🌞

(c) Electricity

$2H_2O \xrightarrow{\text{electricity}} 2H_2 + O_2$ 🔌

(d) Close contact

Iodine and phosphorus react explosively when brought into close contact. 💥

(e) Solution

$NaCl(aq) + AgNO_3(aq) \rightarrow AgCl\downarrow + NaNO_3(aq)$ 💧

(f) Pressure

$HgCl_2 + 2KI \rightarrow HgI_2 + 2KCl$ (when crushed together at high pressure) 🧱

(g) Catalyst

$2H_2O_2 \xrightarrow{MnO_2} 2H_2O + O_2$ ⏩

4. Define :

(a) Photochemical reaction

Reactions that occur with the absorption of light energy are called photochemical reactions. Example: $6CO_2 + 12H_2O \xrightarrow{\text{Sunlight}} C_6H_{12}O_6 + 6O_2 + 6H_2O$. 🌞

(b) Electrochemical reaction

Reactions that are caused or accompanied by the passage of an electric current are known as electrochemical reactions. Example: $2H_2O \xrightarrow{\text{electricity}} 2H_2 + O_2$. ⚡

5. Give an example of each of the following chemical changes.

(a) A photochemical reaction involving

(i) silver salt: $2AgNO_3 \xrightarrow{\text{sunlight}} 2Ag + 2NO_2 + O_2$ 📸

(ii) water: $6CO_2 + 12H_2O \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} C_6H_{12}O_6 + 6O_2 + 6H_2O$ 🌱

(b) A reaction involving

(i) blue solution: $Fe + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu \downarrow$ 🔵

(ii) formation of dirty green precipitate: $FeSO_4(aq) + 2NaOH(aq) \rightarrow Fe(OH)_2 \downarrow + Na_2SO_4(aq)$ 🟢

(c) Two gases combine to form a white solid.

$NH_3(g) + HCl(g) \rightleftharpoons NH_4Cl(s)$ ☁️

(d) Two solids combine to form a liquid.

$C(s) + 2S(s) \xrightarrow{\Delta} CS_2(l)$ (Carbon and sulphur on heating produce liquid carbon disulphide). 💧

(e) A reaction where colour change is noticed.

$Cu(s) + 2H_2SO_4(aq) \rightarrow CuSO_4(aq) + 2H_2O(l) + SO_2(g)$ (Brownish/pink copper turns into blue copper sulphate solution). 🎨

6. Write the chemical reaction where the following changes are observed.

(a) Gas is evolved

$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2 \uparrow$ 💨

(b) Colour change is noticed

$Fe + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu \downarrow$ (Blue solution turns light green). 🎨

(c) Precipitate is formed

$AgNO_3(aq) + NaCl(aq) \rightarrow AgCl \downarrow + NaNO_3(aq)$ (White precipitate). ⬇️

(d) Physical state of reactants is changed

$NH_3(g) + HCl(g) \rightleftharpoons NH_4Cl(s)$ (Gases to solid). 🧊

7. Give reason for the following:

(a) Silver nitrate solution is kept in coloured bottles.

Solutions of silver nitrate are kept in brown bottles in the laboratory because they decompose in the presence of light to form silver, nitrogen dioxide, and oxygen. 🟤

(b) Molybdenum is used in the manufacture of ammonia.

Molybdenum acts as a promoter to improve the efficiency of the iron catalyst during the manufacture of ammonia. 🚀

(c) Blue solution of copper sulphate changes to green when a piece of iron is added to this solution.

Iron is more reactive than copper, so it displaces copper from the solution to form ferrous sulphate ($FeSO_4$), which is light green in colour. 🟢

(d) Colourless concentrated sulphuric acid in a test tube changes to blue on adding a small piece of copper to it.

Copper reacts with concentrated sulphuric acid to form copper sulphate ($CuSO_4$), which gives a blue colour to the solution. 🔵

8. State the changes observed in the following chemical reactions.

(a) A small piece of sodium is dropped in water.

Sodium reacts vigorously with water, darting around the surface and catching fire, forming sodium hydroxide and evolving hydrogen gas. $2Na + 2H_2O \rightarrow 2NaOH + H_2$ 🔥

(b) Few iron filings are added to blue copper sulphate solution.

The blue colour of the solution fades to light green due to the formation of ferrous sulphate, and reddish-brown particles of copper deposit on the iron. $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$ 🎨

(c) Water is added to quick lime (CaO).

A vigorous reaction takes place with a hissing sound, liberating a large amount of heat (boiling the water) and forming a suspension of slaked lime (calcium hydroxide). $CaO + H_2O \rightarrow Ca(OH)_2 +$ Heat ♨️

(d) Hydrochloric acid is added to metal sulphide.

A chemical reaction takes place with the evolution of hydrogen sulphide gas, which has a characteristic offensive, rotten egg smell. $FeS + 2HCl \rightarrow FeCl_2 + H_2S \uparrow$ 👃

1. State the type of reactions and give one equation in each case.

(a) Two or more reactants combine to form a single product.

Combination reaction (or Synthesis). Equation: $C + O_2 \rightarrow CO_2$ ➕

(b) A single reactant breaks up into two or more products.

Decomposition reaction. Equation: $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$ 🪓

2. Give an example of each :

(a) Combination reaction in which :

(i) An element of valency 3 combines with nitrogen.

$2Al + N_2 \rightarrow 2AlN$ (Aluminium combines with nitrogen to form aluminium nitride). ✨

(ii) A non-metal oxide reacting with water to form an acid.

$SO_2 + H_2O \rightarrow H_2SO_3$ (Sulphur dioxide and water form sulphurous acid). 💧

(iii) A metal oxide reacting with water to form its hydroxide.

$CaO + H_2O \rightarrow Ca(OH)_2$ (Calcium oxide reacts with water to form calcium hydroxide). 🧪

(b) Write the decomposition reactions for the following :

(i) Metallic oxide on heating forms metallic oxide and oxygen.

$2PbO_2 \xrightarrow{\Delta} 2PbO + O_2$ 🔥

(ii) A compound on heating gives a gas which turns lime water milky.

$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$ 🍽️

(iii) A white hydroxide on heating gives a yellow residue and water vapours.

$Pb(OH)_2 \xrightarrow{\Delta} PbO + H_2O$ 🟡

3. Write the reaction for the decomposition of a nitrate which does not yield nitrogen dioxide on heating.

$2KNO_3 \xrightarrow{\Delta} 2KNO_2 + O_2$ (Potassium nitrate decomposes to potassium nitrite and oxygen). ♨️

4. Give two balanced equations to show thermal dissociation.

1. $NH_4Cl \rightleftharpoons NH_3 + HCl$ 🔄
2. $N_2O_4 \rightleftharpoons 2NO_2$ 🔄

5. Name :

(a) two coloured oxides that do not decompose on heating.

Iron(III) oxide (Fe_2O_3, brown) and Copper(II) oxide (CuO, black). 🎨

(b) a carbonate that is stable to heat.

Sodium carbonate (Na_2CO_3) or Potassium carbonate (K_2CO_3). 🧱

(c) an oxide formed during thunder and lighting.

Nitric oxide (NO). ⚡

(d) the compound formed when phosphorus and oxygen combines.

Phosphorus pentoxide (P_2O_5). 🍽️

6. Give one example each of :

(a) Thermal decomposition,

$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$ 🔥

(b) Electrolytic decomposition,

$2H_2O \xrightarrow{\text{electricity}} 2H_2 + O_2$ 🔌

(c) Photochemical decomposition.

$2AgCl \xrightarrow{\text{sunlight}} 2Ag + Cl_2$ ☀️

7. Write balanced equations for the following and identify the type of chemical reaction.

(a) Iron reacts with chlorine to form iron (III) chloride.

$2Fe + 3Cl_2 \rightarrow 2FeCl_3$. Type: Direct combination (Synthesis). ➕

(b) Calcium oxide combines with silicon dioxide to form calcium silicate.

$CaO + SiO_2 \rightarrow CaSiO_3$. Type: Direct combination (Synthesis). 🧱

(c) Hydrogen peroxide produces water and oxygen.

$2H_2O_2 \rightarrow 2H_2O + O_2$. Type: Decomposition reaction. 🪓

8. A silvery white metal A is in the form of a ribbon. It burns with a dazzling white flame to form a white powder B. When water is added to powder B, it partially dissolves to form a substance C.

(a) What is metal A ?

Magnesium (Mg). 🎗️

(b) Write the equation for the conversion of A to B and state the type of reaction.

$2Mg + O_2 \rightarrow 2MgO$. Type: Direct combination (Synthesis) or Combustion. 🔥

(c) Write a balanced equation for the conversion of B to C and state the type of reaction.

$MgO + H_2O \rightarrow Mg(OH)_2$. Type: Direct combination (Synthesis). 💧

1. Choose the correct answer from the options given below.

(i) Which of the following is not a characteristic of a chemical change?

(b) No net energy change is involved ❌

(ii) A reaction of a type: $AB + CD \rightarrow AD + CB$, involves:

(c) exchange of ions of AB and CD 🔀

(iii) The reaction: $BaCl_2(aq) + H_2SO_4(aq) \rightarrow BaSO_4(s) + 2HCl(aq)$ is:

(d) double displacement reaction ⬇️

(iv) Thermal decomposition of sodium carbonate will produce:

(d) no other product 🧱

2. Complete the following statements.

(a) The chemical change involving iron and hydrochloric acid illustrates a ..... reaction.

displacement ➡️

(b) In the type of reaction called ..... two compounds exchange their positive and negative radicals.

double decomposition 🔀

(c) A catalyst either ..... or ..... the rate of a chemical change but itself remains ..... at the end of the reaction.

accelerates, retards, unchanged ⏩

(d) On heating, hydrated copper sulphate changes its colour from ..... to ..... .

blue, white 🎨

3. Match the following:

(a) $Zn(s) + H_2SO_4(aq) \rightarrow ZnSO_4(aq) + H_2(g)$ → (iii) Displacement reaction ➡️

(b) $2AgCl(s) \xrightarrow{\text{Sunlight}} 2Ag(s) + Cl_2(g)$ → (i) Photochemical decomposition 🌞

(c) $2KCl \xrightarrow{\text{Electricity}} 2K + Cl_2$ → (iv) Electrolytic decomposition 🔌

(d) $2HgO(s) \xrightarrow{\Delta} 2Hg(s) + O_2(g)$ → (ii) Thermal decomposition 🔥

4. When hydrogen burns in oxygen, water is formed; when electricity is passed through water, hydrogen and oxygen are given out. Name the type of chemical changes involved in the two cases.

- Case 1 (burning): Combination reaction (and Exothermic). ➕
- Case 2 (electricity): Electrolytic decomposition reaction. 🪓

5. Explain the following chemical changes by giving one example of each.

(a) Double decomposition: $NaCl(aq) + AgNO_3(aq) \rightarrow AgCl(s)\downarrow + NaNO_3(aq)$ ⬇️

(b) Thermal dissociation: $NH_4Cl \rightleftharpoons NH_3 + HCl$ 🔄

(c) Reversible reaction: $3Fe + 4H_2O \rightleftharpoons Fe_3O_4 + 4H_2$ 🔁

(d) Displacement: $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$ ➡️

6. (a) What is synthesis? (b) What kind of chemical reaction is synthesis? Support your answer with an example.

(a) Synthesis (or direct combination) is a reaction in which two or more substances combine together to form a single substance. 🤝

(b) It is a combination reaction. Example: Carbon burns in oxygen to form carbon dioxide ($C + O_2 \rightarrow CO_2$). 🔥

7. Decomposition brought about by heat is known as thermal decomposition. What is the difference between thermal dissociation and thermal decomposition.

Thermal decomposition is an irreversible breaking down of a compound into simpler substances by heat. Thermal dissociation is a reversible decomposition reaction brought about only by heat. ⚖️

8. (a) What is a decomposition reaction? (b) Decomposition reactions can occur by (i) heat (ii) electricity and (iii) sunlight. Give two balanced equations for each.

(a) A chemical reaction in which a compound splits into two or more simpler substances is called a decomposition reaction. 🪓

(b)

- **(i) Heat:** $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$ and $2Pb(NO_3)_2 \xrightarrow{\Delta} 2PbO + 4NO_2 + O_2$ ♨️
- **(ii) Electricity:** $2H_2O \xrightarrow{\text{electricity}} 2H_2 + O_2$ and $2NaCl \xrightarrow{\text{electricity}} 2Na + Cl_2$ 🔌
- **(iii) Sunlight:** $2AgCl \xrightarrow{\text{sunlight}} 2Ag + Cl_2$ and $2AgBr \xrightarrow{\text{sunlight}} 2Ag + Br_2$ ☀️

9. Samples of four metals A, B, C and D are taken and were added to the following solutions one by one. The results obtained have been tabulated as follows:

(a) Which is the most reactive metal?

B 🥇

(b) What would you observe when B is added to solution of copper (II) sulphate?

B will displace copper from copper (II) sulphate solution; the blue colour will fade and a reddish-brown deposit will form. 🎨

(c) Arrange the metals A, B, C and D in order of increasing reactivity.

D < C < A < B 📈

10. (a) Define neutralization reaction with an example. (b) Give a balanced equation for this reaction. (c) Give three applications of neutralization reactions.

(a) The reaction between an acid and a base that forms salt and water only. 💧

(b) $NaOH + HCl \rightarrow NaCl + H_2O$ ⚖️

(c) Applications:

- Relieving bee stings using baking soda. 🐝
- Taking antacids (like milk of magnesia) for stomach acidity. 🥛
- Adding slaked lime to acidic soil for agriculture. 🌱

11. What do you understand by precipitation reaction? Explain with an example.

A chemical reaction in which two compounds in their aqueous state react to form an insoluble salt (a precipitate) as one of the products. Example: Barium

chloride reacting with sodium sulphate to form a white precipitate of barium sulphate ($BaCl_2(aq) + Na_2SO_4(aq) \rightarrow BaSO_4(s) \downarrow + 2NaCl(aq)$). ⬇️

12. (a) What are double displacement reactions? (b) Give an example of double displacement reaction, where a gas is evolved.

(a) A type of chemical change in which two compounds in a solution react to form two new compounds by mutual exchange of radicals. 🔀

(b) $FeS(s) + H_2SO_4(aq) \rightarrow FeSO_4(aq) + H_2S(g) \uparrow$. 💨

13. State the type of reactions each of the following equations represent and balance the ones that are not balanced.

(a) $Cl_2 + 2KBr \rightarrow 2KCl + Br_2$

Already balanced. Type: Displacement reaction. ➡️

(b) $NaOH + HCl \rightarrow NaCl + H_2O$

Already balanced. Type: Double decomposition (Neutralization) reaction. 💧

EXERCISE 2(C)

(A) MULTIPLE CHOICE TYPE :

(i) Which of the following is not a chemical change.

Answer: (b) Iron is magnetised 🧲

(ii) Heating of zinc nitrate is a ..... reaction :

Answer: (c) Decomposition 🪓

(iii) Burning of a metal is an example of :

Answer: (c) Direct combination reaction between two elements ➕

(iv) Formation of white precipitate on mixing sodium chloride solution and silver nitrate solution is an example of ..... reaction :

Answer: (d) Double decomposition ⬇️

(v) A brown substance A on heating in air turns black forming another substance B. Substances A and B are :

Answer: (c) A = Cu and B = CuO 🟤

(vi) Which one of these is a chemical change ?

Answer: (c) Combustion of LPG 🔥

(vii) The reaction of neutralisation is a :

Answer: (b) Double decomposition reaction 💧

(viii) $FeS + H_2SO_4 \rightarrow FeSO_4 + H_2S$ **is an example of :**

Answer: (b) Double decomposition reaction 🔀

(ix) Which of the following statements is/are correct with respect to the figure shown alongside? (P - A is HCl, B is iron; Q - A is Cl2, B is copper; R - A is Cl2, B is iron.)

Answer: (c) Only R 🧪

(x) Aman performed the following four experiments. The experiments in which the blue colour of the solution will fade are :

Answer: (c) Q and R 🎨

(xi) The figure given below demonstrates the preparation of oxygen. (I) Which of the following statements is/are correct about the reaction ?

Answer: (b) It is a decomposition reaction and is endothermic. ♨️

(II) Potassium chlorate is :

Answer: (a) an oxidising agent. 🍽️

(III) In the reaction, manganese dioxide :

Answer: (b) acts as a catalyst. ⏩

(xii) What happens when a solution of an acid is mixed with a solution of a base in a test tube ? (P - Temp increases, Q - Temp decreases, R - Temp remains same, S - Salt formation takes place)

Answer: (d) P and S ⚖️

(B) VERY SHORT ANSWER TYPE :

1. Name :

(a) a carbonate which does not decompose on heating.

Answer: Sodium carbonate or Potassium carbonate. 🧱

(b) a nitrate which produces oxygen as the only gas.

Answer: Sodium nitrate or Potassium nitrate. 🧂

(c) a compound which produces carbon dioxide on heating.

Answer: Calcium carbonate. 🔥

(d) a nitrate which produces brown gas on heating.

Answer: Lead nitrate. 🟤

2. Which is the most reactive element in this equation : $CuSO_4 + Fe \rightarrow FeSO_4 + Cu$

Answer: Iron (Fe). ⚡

(C) SHORT ANSWER TYPE :

1. Define : (a) Photochemical reaction (b) Electrochemical reaction. Give one example in each case.

Answer:

- (a) Reactions that occur with the absorption of light energy. Example: $6CO_2 + 12H_2O \xrightarrow{\text{Sunlight}} C_6H_{12}O_6 + 6O_2 + 6H_2O$. 🌞
- (b) Reactions caused or accompanied by the passage of an electric current. Example: $2H_2O \xrightarrow{\text{electricity}} 2H_2 + O_2$. 🔌

2. Give an example of a reaction where the following are involved (a) Evolution of heat (b) Absorption of heat (c) High pressure is required

Answer:

- (a) $CaO + H_2O \rightarrow Ca(OH)_2 +$ Heat 🔥
- (b) $CaCO_3(s) \xrightarrow{\Delta} CaO(s) + CO_2(g)$ ❄️
- (c) $N_2 + 3H_2 \rightleftharpoons 2NH_3$ (above 200 atm). 🧱

3. Why does the blue colour of copper sulphate solution fade on adding iron filings ?

Answer: Iron is more reactive than copper, so it displaces copper from the solution, forming light green ferrous sulphate. 🟢

4. Give an example of : (a) Single displacement reaction (b) Double displacement reaction (c) Photochemical reaction (d) Thermal decomposition reaction (e) Electrolytic decomposition reaction (f) Combination reaction between two compounds

Answer:

- (a) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$ ➡️
- (b) $NaCl + AgNO_3 \rightarrow AgCl \downarrow + NaNO_3$ 🔀
- (c) $2AgCl \xrightarrow{\text{sunlight}} 2Ag + Cl_2$ 📸
- (d) $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$ ♨️
- (e) $2H_2O \xrightarrow{\text{electricity}} 2H_2 + O_2$ ⚡
- (f) $NH_3 + HCl \rightarrow NH_4Cl$ 🤝

5. What do you understand by a 'chemical reaction' ?

Answer: The process of breaking the chemical bonds of reacting substances and making new bonds to form new substances. 🧪

6. Give an example of each of the following chemical changes: (a) A reaction involving (i) change of state (ii) formation of precipitate (b) An exothermic and an endothermic reaction involving carbon as one of the reactants. (c) A reaction where colour change is observed.

Answer:

- (a)(i) $NH_3(g) + HCl(g) \rightleftharpoons NH_4Cl(s)$ ☁️
- (a)(ii) $AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) \downarrow + NaNO_3(aq)$ ⬇️
- (b) Exothermic: $C + O_2 \rightarrow CO_2 + \text{Heat}$. Endothermic: $C + 2S \xrightarrow{\Delta} CS_2$ 🔥❄️
- (c) $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$ (Blue to light green). 🎨

7. What is a chemical change ? Give two examples of chemical changes.

Answer: A permanent change in which the chemical composition of a substance is changed, forming new substances. Examples: Rusting of iron, burning of carbon. 🔄

8. Define exothermic and endothermic reactions. Give two examples in each case.

Answer:

- Exothermic: Reactions that give out heat. (e.g., $C + O_2 \rightarrow CO_2 + \text{Heat}$; $CaO + H_2O \rightarrow Ca(OH)_2 + \text{Heat}$). 🔥
- Endothermic: Reactions that absorb heat. (e.g., $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$; $N_2 + O_2 \xrightarrow{\Delta} 2NO$). ❄️

9. Why is energy involved in a chemical change ?

Answer: Energy is involved because breaking old chemical bonds requires absorption of energy, while forming new bonds releases energy. 🧠

10. A metal X (used for galvanization) is immersed in the aqueous solution of a blue salt YSO4. Metal Y is deposited on the surface of the solution and a strip of metal X dissolves. (a) Predict the metal X. (b) What type of reaction is it ? (c) Write the word equation for this reaction and balance it.

Answer:

- (a) Zinc (Zn). 🛡️
- (b) Single Displacement reaction. ➡️
- (c) Zinc + Copper(II) sulphate → Zinc sulphate + Copper ($Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$). ⚖️

(D) DESCRIPTIVE TYPE :

1. State the main characteristics of chemical reactions. Give at least one example in each case.

Answer:

- Formation of a new substance ($2Na + 2H_2O \rightarrow 2NaOH + H_2$) ✨
- Change of colour ($Fe + CuSO_4 \rightarrow FeSO_4 + Cu$) 🎨

- Evolution of gas ($Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2 \uparrow$) 💨
- Formation of precipitate ($NaCl + AgNO_3 \rightarrow AgCl \downarrow + NaNO_3$) ⬇️

2. State the effects of endothermic and exothermic reactions on the surroundings with examples.

Answer:

- Exothermic reactions release heat, raising the temperature of the surroundings (e.g., burning natural gas). 🔥
- Endothermic reactions absorb heat, lowering the temperature of the surroundings (e.g., dissolving ammonium chloride in water). ❄️

3. Complete and balance the following reactions : (a) $NaCl(aq) + AgNO_3(aq) \rightarrow$ **(b)** $Pb(NO_3)_2 + KI \rightarrow$ **(c)** $CuCO_3 \xrightarrow{\Delta}$ **(d)** $Pb(NO_3)_2 \xrightarrow{\Delta}$ **(e)** $CO_2 + H_2O \xrightarrow{\text{Chlorophyll/Sunlight}}$

Answer:

- (a) $NaCl(aq) + AgNO_3(aq) \rightarrow AgCl(s) \downarrow + NaNO_3(aq)$ ⚖️
- (b) $Pb(NO_3)_2(aq) + 2KI(aq) \rightarrow PbI_2(s) \downarrow + 2KNO_3(aq)$ ⚖️
- (c) $CuCO_3(s) \xrightarrow{\Delta} CuO(s) + CO_2(g) \uparrow$ ⚖️
- (d) $2Pb(NO_3)_2(s) \xrightarrow{\Delta} 2PbO(s) + 4NO_2(g) \uparrow + O_2(g) \uparrow$ ⚖️
- (e) $6CO_2 + 12H_2O \xrightarrow{\text{Chlorophyll/Sunlight}} C_6H_{12}O_6 + 6O_2 + 6H_2O$ ⚖️

4. What do you observe in the following cases ? (a) Lead nitrate is heated. (b) Silver chloride is exposed to sunlight. (c) Hydrogen peroxide is exposed to sunlight. (d) H_2S **gas is passed through copper sulphate solution. (e) Barium chloride solution is added to sodium sulphate solution. (f) Water is added to quicklime. (g) Sodium chloride solution is added to silver nitrate solution.**

Answer:

- (a) Reddish-brown gas (NO_2) evolves, leaving a yellow residue (PbO). 🟤
- (b) White solid turns grey as silver metal forms. 📸
- (c) It decomposes into water and colourless oxygen gas. 💧

- (d) A black precipitate of copper sulphide (CuS) is formed. ⚫
- (e) A white precipitate of barium sulphate ($BaSO_4$) is formed. ⚪
- (f) Vigorous reaction with a hissing sound, boiling water, and releasing heat. ♨️
- (g) A curdy white precipitate of silver chloride ($AgCl$) forms. 🤍

5. Discuss the factors that affect a chemical change.

Answer: Mixing (close contact), solution (aqueous state), heat, light, electricity, pressure, and the presence of a catalyst. ⚙️

6. Balance these reactions and state what type of reactions they are : (a) $NaNO_3 \rightarrow NaNO_2 + O_2$ **(b)** $AgNO_3 + Zn \rightarrow Zn(NO_3)_2 + Ag$ **(c)** $Fe + HCl \rightarrow FeCl_2 + H_2$ **(d)** $AgNO_3 \rightarrow Ag + NO_2 + O_2$ **(e)** $NaBr + Cl_2 \rightarrow NaCl + Br_2$ **(f)** $PbO + C \rightarrow Pb + CO$ **(g)** $KClO_3 \rightarrow KCl + O_2$ **(h)** $AgNO_3 + HCl \rightarrow AgCl + HNO_3$ **(i)** $NH_3 + HCl \rightarrow NH_4Cl$ **(j)** $N_2 + H_2 \rightarrow NH_3$

Answer:

- (a) $2NaNO_3 \rightarrow 2NaNO_2 + O_2$ (Decomposition) 🪓
- (b) $2AgNO_3 + Zn \rightarrow Zn(NO_3)_2 + 2Ag$ (Displacement) ➡️
- (c) $Fe + 2HCl \rightarrow FeCl_2 + H_2$ (Displacement) ➡️
- (d) $2AgNO_3 \rightarrow 2Ag + 2NO_2 + O_2$ (Decomposition) 🪓
- (e) $2NaBr + Cl_2 \rightarrow 2NaCl + Br_2$ (Displacement) ➡️
- (f) $PbO + C \rightarrow Pb + CO$ (Displacement/Redox) ➡️
- (g) $2KClO_3 \rightarrow 2KCl + 3O_2$ (Decomposition) 🪓
- (h) $AgNO_3 + HCl \rightarrow AgCl + HNO_3$ (Double Displacement/Precipitation) ⬇️
- (i) $NH_3 + HCl \rightarrow NH_4Cl$ (Combination) ➕
- (j) $N_2 + 3H_2 \rightleftharpoons 2NH_3$ (Combination) ➕

7. Explain the different types of double decomposition reactions by giving suitable equations.

Answer:

- Precipitation: Forms an insoluble salt. ($NaCl + AgNO_3 \rightarrow AgCl \downarrow + NaNO_3$) ⬇️
- Neutralization: Acid and base form salt and water. ($NaOH + HCl \rightarrow NaCl + H_2O$) 💧

8. Explain thermal dissociation and thermal decomposition reactions with examples. What is the difference between them ?

Answer:

- Thermal decomposition is an irreversible breaking down of a compound by heat. ($CaCO_3 \xrightarrow{\Delta} CaO + CO_2$) 🪓
- Thermal dissociation is a reversible decomposition reaction brought about only by heat. ($NH_4Cl \rightleftharpoons NH_3 + HCl$) 🔁
- Difference: Dissociation is reversible, decomposition is not. ⚖️

9. Metal X was placed in lead nitrate solution. A thin layer of lead metal deposits on metal X. Which is more reactive, metal X or lead ? State the type of reaction and give an example of this type.

Answer: Metal X is more reactive because it displaced lead. This is a single displacement reaction. Example: $Zn + Pb(NO_3)_2 \rightarrow Zn(NO_3)_2 + Pb$. 🥇

10. What type of reaction is respiration ? Explain.

Answer: Respiration is an exothermic decomposition/oxidation reaction. Glucose reacts with oxygen in cells to release energy, carbon dioxide, and water. 🔥

11. Discuss decomposition reactions occurring inside our body.

Answer: Digestion breaks down complex food into simpler substances (e.g., starch into glucose, proteins into amino acids) using enzymes. 🧬

12. Take sodium sulphate solution in a test tube and add barium chloride solution to it dropwise. (a) What do you observe ? (b) Why is the reaction fast ? (c) Which type of reaction is it ?

Answer:

- (a) A white precipitate of barium sulphate forms immediately. ⚪

- (b) It is fast because it involves the rapid exchange of ions in an aqueous solution. ⚡
- (c) Double displacement (Precipitation) reaction. ⬇️

13. $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$. **(a) What will you observe in the above chemical change? (b) Why is this reaction known as a double displacement reaction ? (c) Which reagent can be used to test the chloride radical ?**

Answer:

- (a) A curdy white precipitate forms. 🤍
- (b) Because the two compounds mutually exchange their radicals/ions. 🔀
- (c) Silver nitrate ($AgNO_3$) solution. 🧪

(E) STRUCTURED/APPLICATION/SKILL TYPE :

1. K, Na, Ca, Mg, Al, Zn, Fe, Pb, Cu, [H], Hg, Ag. Pick an element from the reactivity series given above and write the equations for the the following :

(a) Metal hydroxide on heating forms metal oxide and water vapour.

Answer: $Cu(OH)_2 \xrightarrow{\Delta} CuO + H_2O$ 💧

(b) Metal hydroxide on heating forms metal, oxygen and water vapours.

Answer: $4AgOH \xrightarrow{\Delta} 4Ag + O_2 + 2H_2O$ 🥈

(c) Metal nitrate decomposes to give two products only.

Answer: $2KNO_3 \xrightarrow{\Delta} 2KNO_2 + O_2$ ♨️

(d) Metal nitrate on heating forms metal oxide, nitrogen dioxide and oxygen.

Answer: $2Pb(NO_3)_2 \xrightarrow{\Delta} 2PbO + 4NO_2 + O_2$ 🟤

(e) Metal nitrate on heating forms metal, nitrogen dioxide and oxygen.

Answer: $2AgNO_3 \xrightarrow{\Delta} 2Ag + 2NO_2 + O_2$ 🥈

(f) Metal carbonate which is stable to heat.

Answer: Potassium carbonate (K_2CO_3) or Sodium carbonate (Na_2CO_3). 🧱

(g) Metal carbonate which forms metal oxide and carbon dioxide on heating.

Answer: $ZnCO_3 \xrightarrow{\Delta} ZnO + CO_2$ 🍽️

(h) The heating effect on bivalent metal hydrogen carbonate.

Answer: $Ca(HCO_3)_2 \xrightarrow{\Delta} CaCO_3 + H_2O + CO_2$ 🔥

2. $KI + Cl_2 \rightarrow KCl + I_2$. (a) Balance the equation given above. (b) Which type of reaction is it ? (c) Which element is more reactive ? (d) What will be the product formed if KCl reacts with I_2?

Answer:

- (a) $2KI + Cl_2 \rightarrow 2KCl + I_2$ ⚖️
- (b) Single displacement reaction. ➡️
- (c) Chlorine (Cl_2) is more reactive than Iodine (I_2). 🥇
- (d) No reaction will occur because Iodine is less reactive than Chlorine. 🛑

3. See the diagram given below and answer the following questions : (Diagram shows China dish containing copper powder being heated). (a) Write the chemical reaction involved in the process. (b) Mention the colour of : (i) copper powder and (ii) the substance formed after heating it. (c) Can we reverse the above reaction ? If yes, write the equation for the reverse reaction and state the substance that undergoes oxidation and the substance that undergoes reduction.

Answer:

- (a) $2Cu + O_2 \xrightarrow{\Delta} 2CuO$ 🔥
- (b) (i) Reddish-brown (ii) Black 🎨
- (c) Yes, by passing hydrogen gas over hot copper oxide. $CuO + H_2 \xrightarrow{\Delta} Cu + H_2O$. Hydrogen undergoes oxidation; Copper oxide undergoes reduction. 🔄

(F) ASSERTION – REASON TYPE QUESTIONS :

(i) Assertion (A) : Zinc can displace copper from aqueous copper sulphate solution. Reason (R) : Copper is placed above zinc in the reactivity series.

Answer: (c) A is true but R is false. 🥇

(ii) Assertion (A) : In a redox reaction, the electron gaining species acts as a reducing agent. Reason (R) : Oxidation and reduction reactions occur simultaneously in redox reaction.

Answer: (d) A is false but R is true. 🔄

(iii) Assertion (A) : The decomposition of vegetable matter into compost is an endothermic reaction. Reason (R) : Decomposition reactions involve the breakdown of a single reactant into simpler products.

Answer: (d) A is false but R is true. 🌿

(iv) Assertion (A) : The decomposition of silver bromide is utilized in black and white photography. Reason (R) : Light provides energy for this endothermic reaction.

Answer: (a) Both A and R are true and R is the correct explanation of A. 📷

(v) Assertion (A) : Chlorine can displace iodide ions from KI solution. Reason (R) : It is an example of displacement reaction.

Answer: (b) Both A and R are true but R is not the correct explanation of A. 🔁

(vi) Assertion (A) : Silver is displaced by copper from the aqueous solution of silver nitrate. Reason (R) : Copper is placed below silver in the activity series.

Answer: (c) A is true but R is false. 🥈

(G) CASE STUDY :

1. Riya was revising different types of chemical reactions studied in class. She recalled that: A double displacement reaction generally takes place in an aqueous solution where the ions of two compounds exchange places. If one of the products formed is an insoluble solid, the reaction is called a precipitation reaction. A reaction in which an acid or acidic oxide reacts with a base or basic oxide to form salt and water is known as a

neutralisation reaction. To understand these reactions better, Riya observed the following experiments.

(a) When hydrogen sulphide gas (H_2S) was passed through a blue solution of copper sulphate, a black solid was formed along with sulphuric acid. (i) Identify the type of reaction taking place. (ii) Name the black solid formed.

Answer: (i) Double displacement (Precipitation) reaction. (ii) Copper sulphide (CuS). ⚫

(b) Sodium hydroxide reacts with hydrochloric acid to form sodium chloride and water. Explain why this reaction is considered a double displacement reaction.

Answer: Because the Na^+ and H^+ ions from the two reacting compounds mutually exchange places to form two new compounds. 🔀

(c) Complete and balance the following chemical equation : $Pb(NO_3)_2(aq) + KI \rightarrow \ldots . + \ldots .$

Answer: $Pb(NO_3)_2(aq) + 2KI(aq) \rightarrow PbI_2(s) \downarrow + 2KNO_3(aq)$ ⚖️

(d) State one observable change that indicates the reaction in part (c) has taken place.

Answer: The formation of a thick, bright yellow precipitate of lead iodide. 🟡

(H) HOTS QUESTION :

1. An aqueous solution of metal nitrate M reacts with sodium chloride to form a white precipitate of a compound N. N undergoes photochemical reaction (decomposition). Identify and name M and N. Write the chemical equation for the reaction involved and also identify the type of reaction.

Answer:

- M is Silver nitrate ($AgNO_3$). 🧪
- N is Silver chloride ($AgCl$). 🤍
- Equation: $AgNO_3 + NaCl \rightarrow AgCl \downarrow + NaNO_3$. ⚖️
- Reaction type: Double displacement (Precipitation) reaction. ⬇️

2. A water insoluble substance A on reacting with dilute H_2SO_4 releases a gas which turns a paper dipped in acidified potassium dichromate solution green. When the gas is passed through lime water, the solution obtained turns blue litmus red. On bubbling the gas through lime water, it initially became turbid and the turbid nature disappeared when the gas is passed in excess. (a) Identify the substance 'A'. (b) Write the chemical equations of the reactions involved. (c) State the type of reactions that are involved.

Answer:

- (a) Substance '*A*' is an insoluble metal sulphite, such as Calcium sulphite ($CaSO_3$). The gas evolved is Sulphur dioxide (SO_2). 🧿
- (b) Equations:
    1. $CaSO_3 + H_2SO_4 \rightarrow CaSO_4 + H_2O + SO_2 \uparrow$ ⚖️
    2. $Ca(OH)_2 + SO_2 \rightarrow CaSO_3 \downarrow + H_2O$ (Turbidity) ⚖️
    3. $CaSO_3 + H_2O + SO_2 \rightarrow Ca(HSO_3)_2(aq)$ (Turbidity disappears) ⚖️
- (c) The first reaction is a Double displacement reaction. The subsequent reactions are combination/synthesis reactions. 🔀

This is very good! 😊

Here is a comprehensive table detailing the names, chemical formulas, and colours of the compounds and elements mentioned in your chemistry notes: 🎨🧪

Compound / Element Name	**Chemical Formula**	**Colour**
Ammonium dichromate	$(NH_4)_2Cr_2O_7$	Orange
Barium sulphate	$BaSO_4$	White (precipitate)
Bromine (gas)	Br_2	Brown
Chlorine (gas)	Cl_2	Greenish yellow
Chromium oxide	Cr_2O_3	Green solid
Copper (metal)	Cu	Brown red deposit / brownish pink / reddish-brown
Copper carbonate	$CuCO_3$	Green powder
Copper oxide / Copper(II) oxide	CuO	Black
Copper sulphate (hydrated/solution)	$CuSO_4 \cdot 5H_2O$	Blue
Copper sulphate (anhydrous)	$CuSO_4$	White

Compound / Element Name	**Chemical Formula**	**Colour**
Copper sulphide	CuS	Black (precipitate)
Ferrous hydroxide	$Fe(OH)_2$	Dirty green (precipitate)
Ferrous sulphate (crystals/solution)	$FeSO_4 \cdot 7H_2O$	Green / light green
Ferrous sulphate (heated intermediate)	$FeSO_4$	White solid
Iodine (in solution)	I_2	Yellow brown
Iron(III) chloride	$FeCl_3$	Brown
Iron(III) oxide (Ferric oxide)	Fe_2O_3	Brown solid
Lead iodide	PbI_2	Bright yellow / yellow (precipitate)
Lead monoxide	PbO	Yellow residue
Lead nitrate	$Pb(NO_3)_2$	White
Lead sulphate	$PbSO_4$	White
Magnesium oxide	MgO	White powder
Mercuric iodide	HgI_2	Scarlet-coloured

Compound / Element Name	**Chemical Formula**	**Colour**
Nitrogen dioxide (gas)	NO_2	Reddish brown / brown
Potassium iodide	KI	White
Silver (metal)	Ag	Grey metal
Silver bromide	$AgBr$	Light yellow
Silver chloride	$AgCl$	Curdy white (precipitate)
Silver nitrate	$AgNO_3$	Becomes black upon decomposition